Exercise 1

(i) $$\lim_{x\to 0} \frac{1-\sqrt{1-x^2}}{x^2} = \lim_{x\to 0} \frac{(1-\sqrt{1-x^2})(1+\sqrt{1-x^2})}{x^2(1+\sqrt{1-x^2})}$$

$$= \lim_{x\to 0} \frac{\cancel{1}-\cancel{(1}-x^2)}{\cancel{x^2}(1+\sqrt{1-x^2})}$$

$$= \lim_{x\to 0} \frac{1}{1+\sqrt{1-x^2}} = \frac{1}{1+\sqrt{1+0}} = \frac{1}{2}$$

(ii) $$\sin|x| = \begin{cases} \sin(x), & x \geq 1 \\ -\sin(x), & x < 1 \end{cases}$$

Considering left and right hand limits: $f(x) = \frac{\sin|x|}{x}$

$$\lim_{x\to 0^+} \frac{\sin|x|}{x} = \lim_{x\to 0} \frac{\sin(x)}{x} = 1$$

$$\lim_{x\to 0^-} \frac{\sin|x|}{x} = \lim_{x\to 0} (-1)\left[\frac{\sin(x)}{x}\right] = -1$$

$$\lim_{x\to 0^+} f(x) \neq \lim_{x\to 0^-} f(x)$$

$\therefore \lim_{x\to 0} \frac{\sin|x|}{x}$ does not converge

(iii) $$\lim_{x\to 2} \frac{x^2+2x-8}{x(x-2)} = \lim_{x\to 2} \frac{(x+4)\cancel{(x-2)}}{x\cancel{(x-2)}}$$

$$= \lim_{x\to 2} \frac{x+4}{x} = \frac{(2)+4}{(2)} = 3$$

$$\lim_{x\to 1} \frac{x^3-1}{x-1} = \lim_{x\to 1} \frac{\cancel{(x-1)}(x^2+x+1)}{\cancel{(x-1)}} = \lim_{x\to 1} (x^2+x+1)$$

$$= (1)^2+(1)+1 = 3$$

$$\lim_{x\to 0} \frac{x-\frac{1}{x}}{x+\frac{1}{x}} = \lim_{x\to 0} \frac{x^2-1}{x^2+1} = \frac{(0)^2-1}{(0)^2+1} = -1$$

$\lim_{x\to 1} \frac{x^{27}-1}{x-1}$, consider synthetic division:

$$\begin{array}{c|cccccc} (x-1) & 1 & 0 & 0 & 0 & \cdots & -1 \\ & \downarrow & -1 & -1 & -1 & & -1 \\ \hline & 1 & 1 & 1 & 1 & \cdots & 0 \end{array} \Rightarrow \lim_{x\to 1} \frac{x^{27}-1}{x-1} = \lim_{x\to 1} \sum_{r=0}^{26} x^r$$

$$= (1)^{26} + (1)^{25} + \cdots + (1)^0 = 27$$

(iv) $$\lim_{x\to\infty} x\left(\sqrt{x^2+3}-x\right) = \lim_{x\to\infty} \frac{\sqrt{x^2+3}-x}{\frac{1}{x}} = \lim_{x\to\infty} \frac{(x^2+3)-x^2}{\frac{1}{x}(\sqrt{x^2+3}+x)}$$

$$= \lim_{x\to\infty} \frac{3x}{\sqrt{x^2+3}+x} = \lim_{x\to\infty} \frac{3}{\sqrt{1+\frac{3}{x^2}}+1}$$

$$= \frac{3}{\sqrt{1+(0)}+1} = \frac{3}{2}$$

(v) Skip because easy

(vi) $$\lim_{x\to\infty} x^{\frac{2}{3}}\left[(x+1)^{\frac{1}{3}} - (x-1)^{\frac{1}{3}}\right]$$